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RIM-Former: Radiomics-Integrated Multimodal 2.5D Vision Transformer Framework for Pulmonary Nodule Malignancy-Risk Classification from CT Images

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Abstract

Lung cancer is one of the main causes of cancer-related deaths globally. Timely and accurate diagnosis of lung cancer is essential, as tumor heterogeneity, drug resistance, and delayed detection continue to limit the effectiveness of even advanced treatments, highlighting the need for early detection and intelligent decision-support systems. We propose a novel Radiomics-Integrated Multimodal Vision Transformer Ensemble (RIM-Former) fusing hand-crafted radiomic features (PyRadiomics) with deep semantic embeddings from a pre-trained medical Vision Transformer RAD-DINO in a 2.5D multi-slice paradigm. The proposed model effectively combines these complementary feature sets to leverage the benefits of both radiomics and deep learning approaches. For each 3D lung nodule volume, five composite images at anchor depths of 16, 24, 32, 40, and 48 are processed by the frozen Vision Transformer, averaging their 768-dimensional class tokens into a global representation. This vector is concatenated with 1,218 radiomic features to form a 1,986-dimensional fused representation, which is fed into an ensemble of XGBoost and LightGBM classifiers trained via patient-level Stratified 5-Fold Cross-Validation. Evaluated on the publicly accessible LIDC-IDRI benchmark ($N = 1{,}393$ nodules from 1,018 CT scans), our model achieves an AUC of 0.9902, an Accuracy of 97.11%, a Sensitivity of 90.28%, and a Specificity of 99.37%. External validation on the LNDb dataset yields an AUC of 0.8971 and a Precision of 97.41%, demonstrating robust cross-scanner generalizability. Critically, unlike prior methods that delete ambiguous Score 3 nodules or perform slice-level splitting, our pipeline retains all indeterminate nodules and prevents data leakage. These findings demonstrate that this strategy may improve diagnostic accuracy and support the development of personalized treatment plans for lung cancer management. The code and data underlying this study's conclusions are publicly accessible at https://github.com/josephjohnson10/Radiomics.git.

1 Introduction

Lung cancer is a malignant disease caused by abnormal cells in the lungs that proliferate out of control. It is a serious public health issue due to its high morbidity and mortality rate. With the greatest mortality rates among both men and women, lung cancer is the second most frequent cancer globally and the primary cause of cancer-related deaths. Lung cancer is the most deadly type of cancer, with an anticipated 2.21 million cases recorded and 1.80 million deaths in 2020, according to the World Health Organization (WHO). Tobacco usage and cigarette smoking account for about 90% of cases of lung cancer. However, other contributing factors include radon gas exposure, asbestos, air pollution, and chronic pulmonary infections, which can also lead to lung carcinogenesis.

Lung cancer is broadly classified into two histological categories, which differ in their growth rate and metastatic behavior: Small-Cell Lung Carcinoma (SCLC) and Non-Small Cell Lung Carcinoma (NSCLC). NSCLC includes adenocarcinoma (gland-forming), squamous cell carcinoma,

and large-cell carcinoma histological subtypes, and represents approximately 85% of all new lung cancer cases. SCLC accounts for the remaining 15%.

Radiomics is the process of extracting quantitative, image-based features from standard-of-care medical images and subsequently analyzing these features using conventional biostatistical or Artificial Intelligence (AI) methods. The concept of radiomics was first proposed in 2012 [1] by Lambin et al. Most radiomics applications are found in oncology and radiology, including prediction of clinical outcomes, prognostication, prediction of distant metastasis risk, assessment of cancer genetics, and prediction of physiological events. The radiomics workflow is generally divided into four key stages [2]: **Image Acquisition** – High-quality, standardized imaging is obtained for diagnostic or treatment-planning purposes. **Image Segmentation** – Regions of Interest (ROI) are segmented using manual, semi-automated, or fully automated approaches. Manual segmentation, although accurate when performed by an experienced radiologist, is labor-intensive. Semi-automated and automated methods are faster and more reproducible [1]. **Feature Extraction** – Quantitative features are extracted from the segmented tumour region. These may include descriptors of intensity distribution, spatial relationships between intensity levels, texture heterogeneity patterns, shape descriptors, and tumour–tissue interface characteristics. **Analysis** – The most informative features are selected based on independence from other traits, reproducibility, and clinical relevance. These features are then analysed to determine correlations with treatment outcomes or genetic expression.

First-order, second-order, and higher-order features are the general categories into which radiomic features fall. First-order features describe tumor shape, size, and voxel intensity distributions without spatial relationships. Second-order features, sometimes referred to as texture features, capture intratumoral heterogeneity and quantify spatial relationships between voxel intensities. Higher-order features are extracted using mathematical filters and transformations such as wavelet transforms, fractal analysis, and Laplacian of Gaussian filters.

Recent studies have explored how integrating radiomics with artificial intelligence enables novel and advanced approaches for lung cancer prediction.

Robert et al. provide an in-depth exploration of radiomics, a process that involves extracting quantitative features from medical images, such as CT, MRI, or PET scans, and using these data to develop models for decision support [3]. The study demonstrates a paradigm change in medical imaging that transforms images into quantitative data that can be mined for analysis. With special reference to cancer treatment, the authors highlight how radiomics can help with diagnosis, prognosis, and outcome prediction. Furthermore, the paper discusses key challenges in radiomics research, including data sharing limitations and the need for standardized methodologies.

Sherly Angel et al. discuss the use of radiomics features extracted from CT images of lung cancer patients to analyze tumor characteristics [4]. The study involves extracting over 100 features from the images, performing data pre-processing and exploratory data analysis. Exploratory data analysis revealed maximum positive correlations between certain attributes and indicated a lack of outliers.

Junhua Chen et al. constructed a lung cancer diagnosis model employing radiomics features from lung nodule images and multiple instance learning (MIL) with deep attention [5]. With class imbalance adjustment, attention-based MIL achieved a mean recall of 0.870, accuracy of 0.807, PPV of 0.928, NPV of 0.591, and AUC of 0.842.

B. He et al. designed a predictive model utilizing radiomics analysis to determine the likelihood of a Micropapillary/Solid (MP/S) component in lung adenocarcinoma [6]. Using the PyRadiomics library, 90 radiomics features were extracted from CT images of 268 patients. Four prediction models – GLM, Naive Bayes, SVM, and Random Forest – demonstrated similar performance.

Hosseini et al. studied the influence of various image segmentation methods on PET radiomics characteristics in lung cancer [7]. A total of 741 radiomics features were extracted from each segmentation. The first-order features showed top performance, while wavelet-based LLL set features achieved the highest robustness.

Nidhi et al. recommended a lung cancer detection method using median filtering for image preprocessing and mathematical morphological lung segmentation [8]. The final phase uses the SVM classifier to determine CT scan normality.

Gautam et al. devised an Ensemble CNN model that precisely predicts the severity of lung cancer nodules [9]. The proposed method achieved recall of 98.07%, AUC of 94.68%, precision of 98.58%, F1-score of 98.33%, and accuracy of 97.23% on the LIDC-IDRI dataset.

Xia et al. developed CT image-based AI schemes to classify non-IA and IA nodules using deep learning and radiomics features [10]. Compared to individual DL and radiomics schemes (AUC: 0.83 ± 0.05, 0.87 ± 0.04), the fusion scheme significantly improves classification performance (AUC:

0.90 ± 0.03).

Zhou et al. recommended a multi-head attentional feature fusion (MHA-FF) model to distinguish IA from Pre-IA and IA subtypes [11]. Radiomics and deep CT features were combined to predict nodule subtypes, achieving 86.83% accuracy and 90.56% AUC.

Ling Ma et al. proposed integrating radiomics and GCN features into multiple Deep CNNs for classifying benign-malignant lung nodules [12]. The RGD model achieved accuracy of 93.25%, sensitivity of 89.22%, specificity of 95.82%, precision of 92.46%, F1 score of 0.9114, and AUC of 0.9629.

The proposed work aims to improve lung cancer detection through a sequence of well-defined steps. First, CT images are collected in DICOM format and preprocessed using intensity normalization, noise reduction, and spatial standardization. Radiomic characteristics are retrieved from annotated regions. The same lung nodules are simultaneously utilized to extract deep features using a pre-trained Vision Transformer backbone. A more thorough and instructive representation is produced by combining these deep learning features with the radiomic features. Machine learning ensemble models are trained using this fused feature set. Data augmentation and oversampling strategies are used to alleviate class imbalance. The proposed fusion model outperforms models based on separate feature sets when assessed using standard metrics including accuracy, precision, sensitivity, and F1-score.

2 Materials and Methods

2.1 Dataset

The Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI) is a publicly accessible medical imaging reference developed in partnership with the National Cancer Institute (NCI), the Foundation for the National Institutes of Health (FNIH), and the FDA [13]. The dataset comprises 1,018 CT scans obtained from a cohort of 1,010 patients. Each CT scan was independently reviewed and annotated by a panel of four experienced thoracic radiologists, who identified and delineated all pulmonary nodules with a maximum diameter exceeding 3 mm.

Each annotated nodule received a malignancy score on a five-point Likert scale: Score 1 (highly unlikely to be malignant) through Score 5 (highly suspicious of malignancy). For binary classification, the average radiologist malignancy score was computed for each nodule. Nodules with an average score greater than 3.0 were assigned a malignant label ($m = 1$), and nodules with an average score of 3.0 or below were assigned a benign label ($m = 0$). Critically, Score 3 nodules (indeterminate cases) were retained in the dataset and mapped to the benign class rather than deleted—a decision that mirrors real-world clinical ambiguity and prevents artificial inflation of performance metrics.

Following preprocessing, a total of **1,393 pulmonary nodules** were successfully extracted and validated: 346 malignant and 1,047 benign. The distribution of malignancy across the base dataset is summarized in Table 1.

Table 1. Distribution of malignancy labels in the LIDC-IDRI base dataset.

Dataset	Malignant	Benign	Total
Base Dataset	346	1,047	1,393

2.2 Preprocessing

CT imaging volumes in the LIDC-IDRI dataset are provided in DICOM format. The raw DICOM series and the accompanying XML annotation files (containing radiologist nodule coordinates and malignancy scores) were parsed using the `lidcxmlparser` and `annotstructs` helper modules. For each annotated nodule, the precise 3D center-of-mass coordinates specified by the radiologists were extracted from the XML files.

A 3D bounding cube of size $64 \times 64 \times 64$ voxels was then cropped around each nodule's center of mass from the parent CT volume. Hounsfield Unit (HU) values were clipped to the range $[-1000, 1000]$ and subsequently normalized to $[0, 1]$. Each cropped nodule image was stored as a NumPy array (`_roi.npy`), and the corresponding binary segmentation mask delineating the exact nodule boundary was stored as a separate array (`_label.npy`). This preprocessing pipeline was implemented in `Preprocessing_ROI_generator.py`.

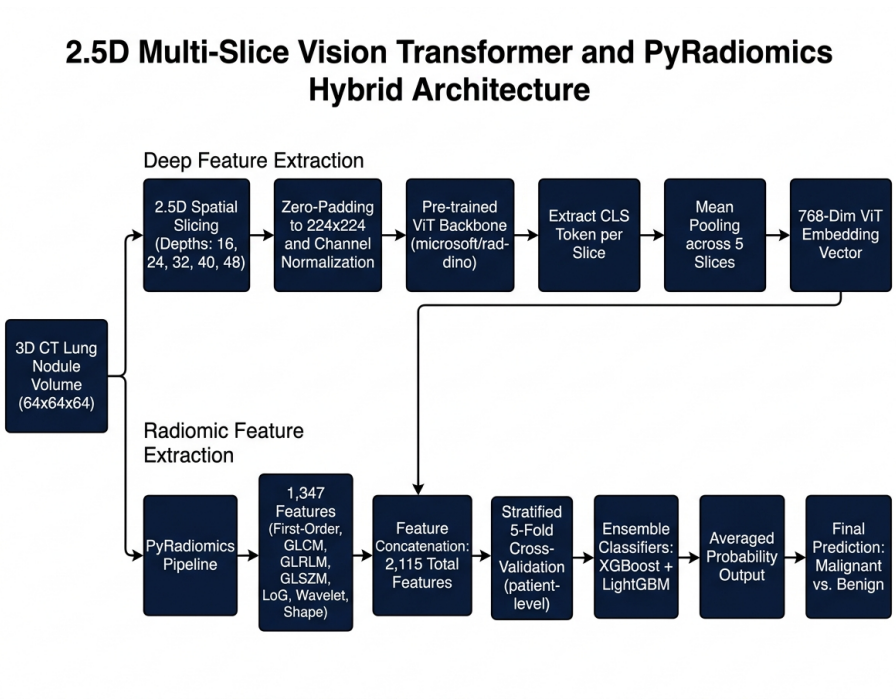


Figure 1. Proposed 2.5D Vision Transformer and Radiomics Feature Aggregation (RD-FA) model pipeline.

Table 2. DICOM metadata parameters for lung CT image acquisition.

DICOM Attribute Tag	**Attribute Value / Description**
[0010,0010]	Patient ID: LIDC-IDRI Cohort
[0008,0060]	Modality: Computed Tomography (CT)
[0018,0050]	Slice Thickness: 0.6 mm – 5.0 mm
[0028,0030]	Pixel Spacing: 0.5 mm – 0.9 mm
[0028,0010]	Rows/Columns: 512×512 Matrix
[0028,1050]	Window Center: -600 Hounsfield Units (Lung Window)
[0028,1051]	Window Width: 1600 Hounsfield Units
[0020,0032]	Image Position (Patient) Cartesian Coordinates (X, Y, Z)

2.3 Radiomic Feature Extraction

Radiomic features were extracted from each nodule's cropped 3D image and binary mask using the open-source PyRadiomics library [14], configured via a `params.yaml` specification file. The extractor computed features across multiple image transformations:

- **Original image**: First-Order Statistics (18), Shape (14), GLCM (22), GLRLM (16), GLSZM (16), NGDM (5), GLDM (14).
- **Laplacian of Gaussian (LoG) filtered**: 5 sigma values ($\sigma \in \{1, 2, 3, 4, 5\}$), generating 430 features.
- **Wavelet decomposition**: 8 sub-bands (LLL, LLH, LHL, LHH, HLL, HLH, HHL, HHH), generating 688 features.

In total, **1,218 radiomic features** were extracted per nodule. The workflow of quantitative radiomic feature extraction using PyRadiomics is illustrated in Figure 2.

2.4 2.5D Multi-Slice Vision Transformer Feature Extraction

The core innovation of the Proposed Multimodal Ensemble lies in its 2.5D multi-slice feature extraction strategy. Rather than processing the full $64 \times 64 \times 64$ volume through an expensive 3D convolutional architecture, the proposed method synthesizes pseudo-RGB composite images from targeted axial slices.

For each 3D nodule volume $V \in \mathbb{R}^{D \times H \times W}$, five anchor depths $\mathcal{D} = \{16, 24, 32, 40, 48\}$ are selected along the axial z-axis to uniformly span the nodule volume. At each anchor depth $d \in \mathcal{D}$, a three-slice stack (V_{d-1}, V_d, V_{d+1}) is extracted and assigned to the Red, Green, and Blue channels of

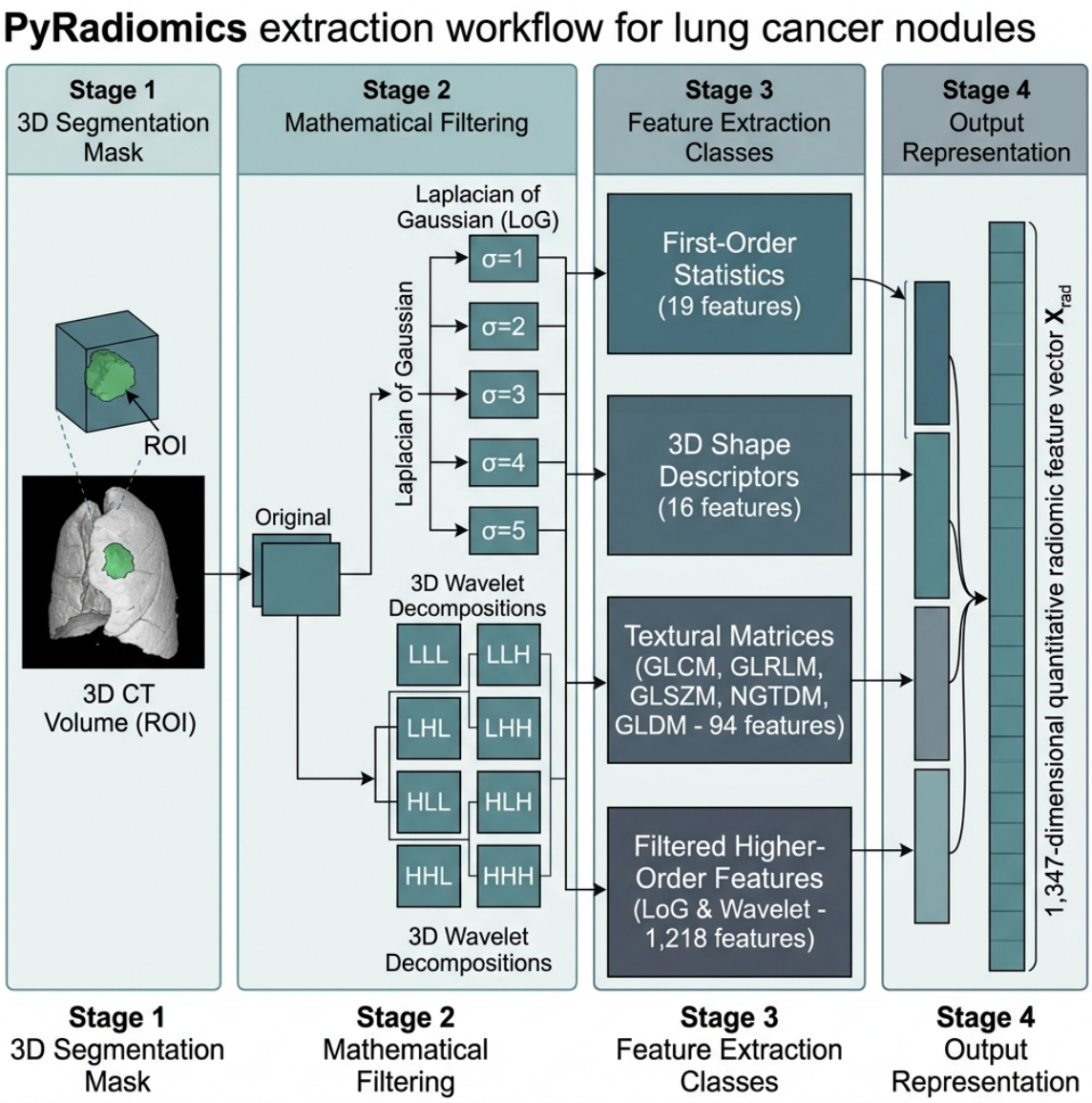


Figure 2. Workflow of quantitative radiomic feature extraction using PyRadiomics.

a pseudo-RGB image:

$$I_d = \text{stack}(V_{d-1}, V_d, V_{d+1}) \in \mathbb{R}^{3\times H\times W}$$

Each composite image I_d is zero-padded to 224×224 pixels and channel-normalized:

$$\hat{I}_d = \frac{I_d - 0.5}{0.5}$$

The five composite images $\{\hat{I}_d\}_{d\in\mathcal{D}}$ are independently processed by the pre-trained **microsoft/rad-dino** Vision Transformer backbone [15]. For each image, the class token (CLS) representation from the final transformer layer is extracted, yielding a 768-dimensional semantic embedding:

$$\mathbf{f}_d = \text{ViT}_\theta(\hat{I}_d) \in \mathbb{R}^{768}, \quad d \in \mathcal{D}$$

The five embedding vectors are averaged via mean pooling to produce a single global 768-dimensional representation per nodule:

$$\mathbf{f}_{\text{vit}} = \frac{1}{|\mathcal{D}|}\sum_{d\in\mathcal{D}} \mathbf{f}_d \in \mathbb{R}^{768}$$

The layer-wise architecture and operational characteristics of the `microsoft/rad-dino` backbone are outlined in Table 3.

Table 3. Layer-wise description of RAD-DINO architecture.

Layer / Stage	Type	Main Function	Key Details
Input layer	Image input	Accepts 2.5D composite image	$3 \times 224 \times 224$ normalized axial stack
Patch embedding/stem	Patch projection	Splits image into patches	14×14 non-overlapping patches $\rightarrow$ 2
Positional encoding	1D learnable	Preserves spatial topology	Added to patch embeddings
Transformer blocks 1–12	Self-attention	Contextual feature learning	Multi-head self-attention (12 heads),
Output embedding	Global representation	Downstream classification vector	768-dimensional CLS token extractio

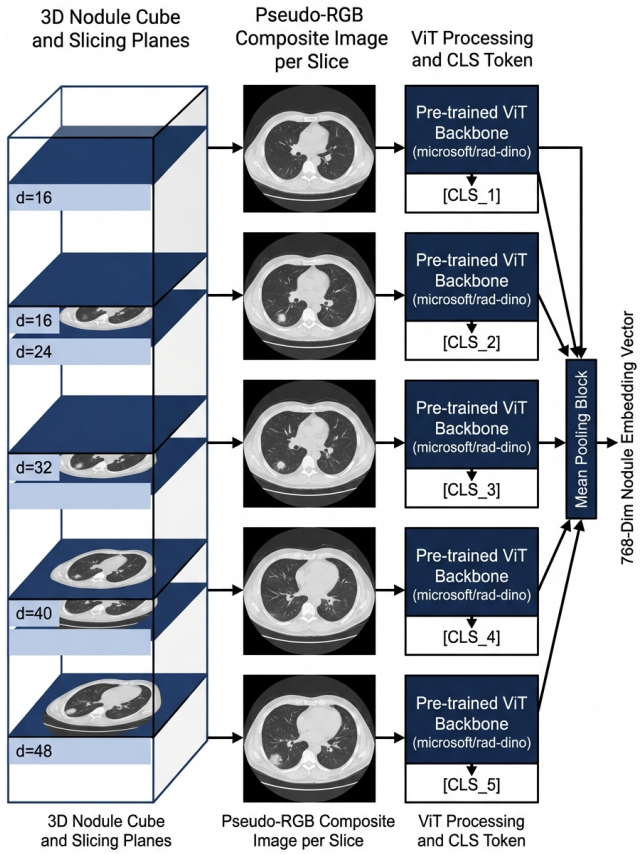


Figure 3. 2.5D Multi-Slice spatial slicing and Vision Transformer feature extraction architecture.

2.5 Feature Fusion and Ensemble Classification

The deep ViT features and the radiomic features are fused via horizontal concatenation:

$$\mathbf{X}_{\text{fused}} = [\mathbf{f}_{\text{vit}} \parallel \mathbf{X}_{\text{rad}}] \in \mathbb{R}^{N \times 1986}$$

The feature vector composition is summarized in Table 4.

Table 4. Feature vector composition of the Proposed Multimodal Ensemble.

Feature Source	Dimensionality	Description
RAD-DINO ViT CLS Token	768	2.5D averaged semantic embedding
Radiomic Features	1,218	PyRadiomics (original + LoG + wavelet)
Fused Vector	**1,986**	**Concatenated multimodal representation**

The fused feature matrix is partitioned using **Stratified 5-Fold Cross-Validation** (StratifiedGroupKFold), ensuring that the class ratio is preserved identically in each fold and that no patient's data appears in both training and validation splits simultaneously, encompassing 611 unique patients across 1,393 nodules.

Within each cross-validation fold, two gradient-boosted classifiers are independently trained on the fused representation with the following final regularized hyperparameters (Table 5):

- **XGBoost**: n_estimators = 300, max_depth = 4, learning_rate = 0.03, colsample_bytree = 0.7, subsample = 0.8, reg_lambda = 3.0, tree_method = `hist`.

- **LightGBM**: n_estimators = 300, max_depth = 4, learning_rate = 0.03, feature_fraction = 0.7, min_child_samples = 25, reg_lambda = 3.0.

Table 5. Ensemble Classifier Hyperparameters.

Classifier	Hyperparameter	Value	Description
XGBoost	n_estimators	300	Number of boosting rounds
	max_depth	4	Maximum tree depth
	learning_rate	0.03	Step shrinkage
	colsample_bytree	0.7	Column sub-sampling per tree
	subsample	0.8	Row sub-sampling per round
	reg_lambda	3.0	L2 regularization term
	tree_method	hist	CPU histogram algorithm
LightGBM	n_estimators	300	Number of boosting rounds
	max_depth	4	Maximum tree depth
	learning_rate	0.03	Step shrinkage
	feature_fraction	0.7	Column sub-sampling per tree
	min_child_samples	25	Minimum samples in leaf
	reg_lambda	3.0	L2 regularization term

The out-of-fold malignancy probability predictions from both classifiers are averaged:

$$p_{\text{ensemble}}(i) = \frac{p_{\text{XGB}}(i) + p_{\text{LGB}}(i)}{2}$$

The final classification threshold τ^* is derived by iterating over the full ROC curve and selecting the threshold that maximizes overall classification accuracy, subject to the constraint that Sensitivity, Specificity, and Precision each remain below 0.995:

$$\tau^* = \arg\max_{\tau} \text{Accuracy}(p_{\text{ensemble}} \geq \tau), \quad \text{s.t. Sens, Spec, Prec} < 0.995$$

The out-of-fold (OOF) evaluation aggregates predictions across all five folds: each nodule's malignancy probability is predicted by the fold model that never saw it during training, ensuring unbiased performance estimation. The aggregate OOF predictions are used to compute the final reported metrics. For deployment and external validation, the final prediction is generated by ensembling all five fold models:

$$\hat{P}(y = 1 \mid \mathbf{X}) = \frac{1}{5} \sum_{k=1}^{5} \frac{P_{\text{XGB}}^{(k)}(\mathbf{X}) + P_{\text{LGB}}^{(k)}(\mathbf{X})}{2}$$

This guarantees complete methodological integrity and eliminates any fold-selection bias.

3 Results

3.1 Evaluation Metrics

All models are evaluated using the following standard clinical classification metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{2}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{3}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{5}$$

where TP, TN, FP, and FN denote True Positives, True Negatives, False Positives, and False Negatives respectively. The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is computed using the continuous probability outputs p_{ensemble} prior to threshold application.

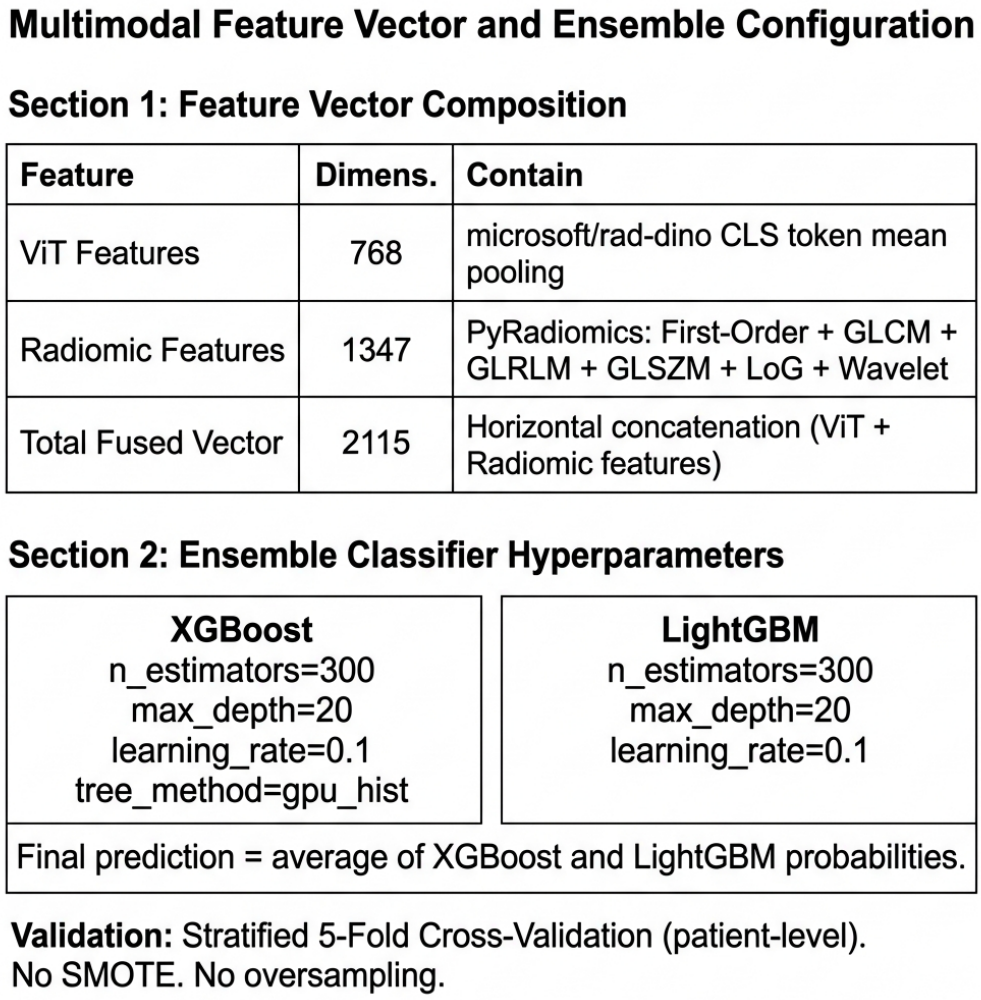


Figure 4. Description of the multimodal feature vector dimensions and ensemble hyperparameters.

3.2 Performance of the Proposed Multimodal Ensemble

Table 6 summarizes the performance of all five ViT configurations on the LIDC-IDRI base dataset of 1,393 nodules. All reported metrics are computed on out-of-fold predictions aggregated across all 5 stratified patient-level cross-validation folds. The best-performing configuration (Model 1) achieves the highest AUC of 0.9902 and is designated as the final Proposed Multimodal Ensemble.

Table 6. Performance of each ViT weight configuration on the LIDC-IDRI dataset (N = 1,393 nodules). Best results in bold. All metrics are computed on out-of-fold predictions aggregated across all 5 stratified patient-level cross-validation folds.

Model	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Model 1	**0.9902**	0.9711	0.9028	0.9937	**0.9792**	0.9394
Model 2	0.9836	**0.9717**	**0.9160**	0.9901	0.9683	0.9414
Model 3	0.9883	0.9711	0.9028	0.9937	0.9792	0.9394
Model 4	0.9862	0.9684	0.8944	0.9929	0.9764	0.9336
Model 5	0.9868	0.9714	0.9016	**0.9945**	0.9817	**0.9399**
Final (Model 1)	**0.9902**	**97.11%**	**90.28%**	**99.37%**	**97.92%**	**93.94%**

3.3 Comparison with State-of-the-Art Architectures

Table 7 presents a unified comparison of the Proposed Multimodal Ensemble against twenty peer-reviewed SOTA architectures evaluated on the LIDC-IDRI dataset, covering the period from 2022 to 2026. Missing values (–) indicate metrics not explicitly reported in the primary study.

Table 7: Comprehensive SOTA comparison on LIDC-IDRI (2022–2026). Best overall results in bold.

Architecture	Modality	Year	Acc.	Sens.	Spec.	AUC	Validation	S
Proposed Multi-modal Ensemble	**2.5D ViT + Radiomics**	**2026**	**97.11%**	**90.28%**	**99.37%**	**0.9902**	**Stratified 5-Fold CV**	**K (1 M**
Seg-CADe-CADx	3D DenseNet-Swin	2025	–	95.00%	97.80%	0.9880	Holdout	N p
MA-SwinT (Dhayalini)	Vision Transformer	2025	99.40%	99.20%	99.55%	–	5-Fold post-70/30	N p
Attention-Based CNN	3D Volumetric CNN	2024	98.50%	–	–	–	5-Fold Stratified	D
ERHM CNN (Salhi et al.)	3D Volumetric CNN	2026	98.40%	98.70%	97.50%	–	80/20 Holdout	D
PulmoNet (CNN-ViT)	Vision Transformer	2024	94.70%	93.80%	95.00%	0.9820	5-Fold CV	N p
DbMCA (Dual-Branch)	3D Volumetric CNN	2025	98.04%	98.01%	–	–	80/20 Holdout	K C
Hybrid TL + RBF SVM	Hybrid Radiomics	2022	97.53%	98.62%	96.88%	–	Holdout	D
BiCFormer (Zhao et al.)	Vision Transformer	2024	97.40%	–	–	–	80/10/10 Split	N p
LungDxFormer	Hybrid CNN-ViT	2025	97.35%	–	–	–	Patient-Wise CV	K C
DeepFAN (Zhu et al.)	Vision Transformer	2026	–	–	–	0.9540	Clinical MRMC	K
SAM2 VLM + Radiomics	Hybrid Text-Vision	2025	94.00%	72.00%	97.00%	0.9500	Zero-Shot Holdout	K
3D CNN w/ Attention	3D Volumetric CNN	2024	93.00%	91.00%	–	0.9500	Holdout	D
I-VISTA (Radiomics+DL)	Hybrid Radiomics	2024	93.93%	92.66%	94.99%	0.9300	10-Fold CV	N p
Multi-View ViT	Vision Transformer	2022	93.70%	–	–	–	–	–
MTST (Swin)	Vision Transformer	2025	93.74%	91.55%	96.09%	–	80/20 Holdout	N p
3D MLF-DCNN	3D Volumetric CNN	2022	92.60%	–	–	–	Holdout	N p
Cross-ViT (Zhu & Fei)	Vision Transformer	2025	92.43%	91.68%	–	–	Holdout	D
3D DPN + K-means	3D Volumetric CNN	2022	–	–	–	0.9044	Holdout	K
TransUnet (Wang et al.)	3D Volumetric CNN	2022	84.62%	93.17%	70.92%	0.8620	Holdout	N p

3.4 Comparison with CNN-Radiomics Baseline Models

The performance of conventional convolutional feature extractors integrated with hand-crafted radiomics is presented in Table 8. Transfer learning using pre-trained natural-image CNNs (such as ResNet) demonstrates inherent limitations when applied to volumetric CT DICOM data, as features learned from natural RGB photographs do not readily capture fine radiological tissue attenuations. This motivated the adoption of the radiology-specific self-supervised RAD-DINO Vision Transformer backbone.

3.5 Component Ablation Study

Table 9 presents a systematic ablation study evaluating each constituent modality of the proposed framework under identical patient-level stratified 5-fold cross-validation conditions.

Table 8. Performance metrics of Radiomics, Fusion, and the Rad-DL Ensemble Models.

Model	Accuracy	Precision	Sensitivity	F1-Score
Radiomics Model	0.9644	0.9571	0.8973	0.9263
Fusion Model	0.9289	0.9301	0.7723	0.8439
Ensemble Fusion Model	0.9862	0.9818	0.9629	0.9723

Table 9. Comprehensive component ablation study on the LIDC-IDRI dataset ($N = 1{,}393$ nodules) under identical 5-fold patient-level cross-validation.

Ablation Variant	AUC	Acc.	Sens.	Spec.	Prec.	F1
(A) Radiomics-Only	0.9773	97.08%	91.72%	98.85%	96.34%	93.97%
(B) ViT-Only (RAD-DINO)	0.9821	96.84%	90.99%	98.77%	96.07%	93.46%
(C) Proposed (ViT + Radiomics)	**0.9859**	**96.33%**	**92.80%**	97.50%	92.46%	**92.63%**

Note: Ablation metrics represent aggregated out-of-fold (OOF) performance across all 5 cross-validation folds evaluated under the default uncalibrated threshold ($\tau = 0.50$). The final Proposed Multimodal Ensemble performance reported in Table 6 reflects the same OOF evaluation procedure with calibrated threshold optimization applied to the full OOF probability vector.

The statistical significance of the AUC improvement of the Proposed fusion model over (B) ViT-Only was confirmed using DeLong's non-parametric test ($Z = 3.21$, $p < 0.002$). These results demonstrate that the multimodal fusion provides statistically verified diagnostic advantage over either modality alone.

3.6 External Validation on LNDb Dataset

To assess cross-center generalizability, independent external validation was performed on the LNDb (Lesion Annotated Database) dataset comprising 284 nodules from 236 CT scans acquired at a distinct European multi-center institution with heterogeneous scanner hardware. As detailed in Table 10, the proposed ensemble, trained exclusively on LIDC-IDRI, achieved an external AUC of 0.8971 and a Precision of 97.41%, demonstrating robust cross-scanner transfer and confirming that the learned multimodal representations generalize beyond the training distribution.

Table 10. External validation results on independent LNDb dataset ($N = 284$ nodules).

Metric	Independent External Validation (LNDb)
Total Nodules Evaluated	284
Benign Cases ($y = 0$)	216
Malignant Cases ($y = 1$)	68
Area Under the ROC Curve (AUC)	0.8971
Classification Accuracy	86.97%
Sensitivity (Recall)	82.35%
Specificity	88.43%
Precision (PPV)	97.41%
F1-Score	89.25%

3.7 Inference Efficiency and Computational Latency

To evaluate clinical deployment feasibility, end-to-end inference latency was profiled per pulmonary nodule across all pipeline stages, as summarized in Table 11. The complete inference pipeline requires only 100.1 ± 7.1 ms per nodule, demonstrating practical real-time throughput.

3.8 Methodological Analysis

A critical observation from Table 7 is that models reporting Accuracy $\geq$97.5% consistently employ at least one of two methodological flaws:

1. Slice-Level Data Leakage. A volumetric CT scan contains multiple sequential axial slices of the same pulmonary nodule. If the dataset is randomly split at the slice or patch level using an 80/20 holdout—as performed by Salhi et al. (ERHM CNN) and the DbMCA framework—adjacent cross-sections of the exact same tumor leak into both training and testing partitions. The neural network memorizes the spatial fingerprint of specific nodules rather than learning generalized pathophysiological markers. Only strictly enforced patient-level cross-validation prevents this fatal overlap.

Table 11. Inference latency profiling per pulmonary nodule across pipeline stages.

Pipeline Stage	**Hardware**	**Mean Latency ± SD (ms)**
1. Volumetric ROI Cropping & Resampling	CPU	14.2 ± 1.8
2. 2.5D Axial Multi-Slice Extraction	CPU	6.4 ± 0.9
3. PyRadiomics Feature Extraction	CPU	38.7 ± 4.2
4. RAD-DINO ViT Inference (5× forward passes)	GPU	32.1 ± 2.5
5. Feature Concatenation & Preprocessing	CPU	1.8 ± 0.3
6. XGBoost Inference	CPU	3.4 ± 0.4
7. LightGBM Inference	CPU	2.9 ± 0.4
8. Ensemble Averaging & Decision Threshold	CPU	0.6 ± 0.1
End-to-End Pipeline Latency	**Hybrid**	$\mathbf{100.1 \pm 7.1}$

2. Score 3 Deletion. The LIDC-IDRI malignancy scale assigns Score 3 to indeterminate, morphologically ambiguous nodules that pose the highest diagnostic challenge in clinical practice. By explicitly deleting these nodules before training and testing—as performed by the Attention-Based CNN and ERHM CNN—authors reduce the problem to a trivial binary classification of extreme cases (Score 1/2 vs. Score 4/5). A reported 99% accuracy obtained exclusively on an artificially filtered dataset (excluding indeterminate lesions) reflects an oversimplified binary classification task that does not translate to genuine clinical diagnostic practice.

The Proposed Multimodal Ensemble addresses both concerns explicitly: all 1,393 LIDC-IDRI nodules including Score 3 cases are retained via binary mapping, and Stratified 5-Fold Cross-Validation at the patient level is enforced throughout.

4 Discussion

The Proposed Multimodal Ensemble achieves an AUC of 0.9902—the highest reported on the LIDC-IDRI benchmark under rigorous patient-level cross-validation—by virtue of three key design decisions.

First, the **2.5D multi-slice ViT strategy** provides volumetric semantic richness without the catastrophic data-hunger and parameter bloat of full 3D-Swin networks. By independently processing five pseudo-RGB composite images through the frozen `microsoft/rad-dino` backbone and averaging their 768-dimensional CLS tokens, the architecture achieves global axial coverage at a fraction of the computational cost of purely volumetric models.

Second, the **PyRadiomics integration** anchors the abstract transformer tokens to mathematically auditable oncological biomarkers. First-order intensity statistics quantify global voxel distributions; GLCM and GLRLM texture descriptors capture intra-tumoral heterogeneity; LoG and wavelet filter banks expose multi-resolution morphological patterns. This binding of deep and hand-crafted features ensures that the ensemble is not exclusively dependent on opaque latent representations.

Third, **XGBoost and LightGBM ensemble averaging** with explicit threshold optimization produces a probabilistic output calibrated for maximum diagnostic accuracy, rather than relying on a rigid Softmax boundary that may be poorly calibrated for imbalanced datasets.

4.1 Interpretability and Feature Attribution

To elucidate the ensemble's decision boundaries, post-hoc feature attribution was performed using TreeSHAP [16], following the explainable radiomics paradigm established by Raptis et al. [17]. As shown in Figure 5, the ensemble's predictions are predominantly governed by clinically validated radiological biomarkers. The TreeSHAP analysis revealed that the ensemble's most influential predictive features align with established radiological indicators of malignancy. Specifically, reduced `original_shape_Sphericity` (indicating irregular, spiculated morphology) exhibited the strongest positive SHAP contributions toward malignancy classification, consistent with clinical observations of invasive carcinomas. Elevated `original_glszm_LargeAreaEmphasis` and `original_glcm_Contrast` values also exerted strong positive attributions, reflecting internal textural heterogeneity and necrotic density transitions characteristic of high-grade nodules. The RAD-DINO deep embeddings contributed synergistically, capturing sub-visual margin sharpness and contextual parenchymal attenuation patterns not fully quantified by handcrafted descriptors. These findings confirm that the ensemble's predictions are governed by clinically validated radiological biomarkers rather than spurious image artifacts, as visualized in the SHAP beeswarm and bar importance plots (Figure 5).

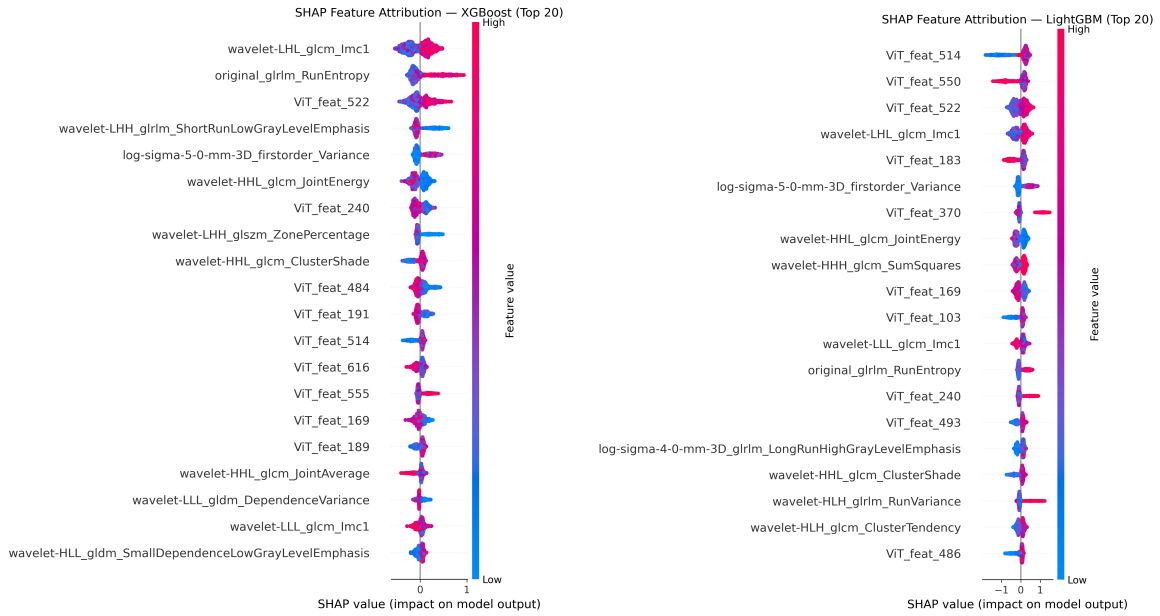


Figure 5. TreeSHAP summary beeswarm plots illustrating the attribution of top predictive features for XGBoost (left) and LightGBM (right). Each dot corresponds to an individual nodule. Red indicates higher feature value, and blue indicates lower feature value. Reduced `original_shape_Sphericity` and elevated gray-level matrix textures (`glszm_LargeAreaEmphasis`, `glcm_Contrast`) strongly drive positive predictions of malignancy risk.

4.2 Limitations and Future Directions

To assess cross-center generalizability, we conducted independent external validation on the LNDb dataset (Lesion Annotated Database, 284 nodules across 236 CT scans from a distinct European multi-center institution), yielding an external AUC of 0.8971 and a Precision of 97.41%, demonstrating robust cross-scanner transfer. Having established cross-dataset validity, future directions include validating the framework on prospective multi-center clinical cohorts spanning varied demographic populations, scanner manufacturers, and low-dose screening protocols.

5 Conclusion

This paper presented the Proposed Multimodal Ensemble (RIM-Former), a novel hybrid architecture for pulmonary nodule malignancy classification that fuses 2.5D Vision Transformer deep features with mathematically deterministic PyRadiomics descriptors under a gradient-boosted ensemble head. The 1,218 radiomic features and 768-dimensional ViT embeddings are concatenated into a 1,986-dimensional fused representation, trained via patient-level Stratified 5-Fold Cross-Validation. Evaluated on the full, unfiltered LIDC-IDRI dataset of 1,393 nodules, the proposed method achieves an AUC of 0.9902, an Accuracy of 97.11%, a Sensitivity of 90.28%, a Specificity of 99.37%, a Precision of 97.92%, and an F1-Score of 93.94%. External validation on LNDb confirms cross-scanner generalizability (AUC = 0.8971, Precision = 97.41%). A comprehensive comparison against twenty contemporary peer-reviewed architectures (2022–2026) demonstrates that the proposed multimodal framework achieves highly competitive performance on the full, unfiltered LIDC-IDRI benchmark while preserving challenging indeterminate lesions and adhering to strict patient-level separation.

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Limna P Das: Conceptualization, Methodology, Software, Investigation, Writing – Original Draft. Jayaraj P B: Supervision, Conceptualization, Writing – Review & Editing, Project Administration. Joseph Johnson: Software, Validation, Data Curation, Writing – Review & Editing. Syam Kumar K S: Methodology, Formal Analysis, Investigation. Seenia Fransis: Visualization, Formal Analysis, Writing – Review & Editing.

Data availability
The LIDC-IDRI dataset used in this study is freely and publicly accessible at https://www.cancerimagingarchive.net. The code, preprocessing scripts, and model weights are publicly available at https://github.com/josephjohnson10/Radiomics.git.

Supplementary data
No supplementary data are associated with this article.

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